2026 Exam SBA

Description of the Project

The citizens of Demlinbe, a country in the English-speaking Caribbean, are unable to purchase housing for their families, mainly due to the extremely high cost of real estate and building materials. Middle and low income earners are severely affected. The government, through its agency, the Authority of Housing Ministry, has intervened however, and has embarked on an extensive house building exercise in three selected parts of the country.

Citizens were invited to apply for the chance to own one of these homes. You are asked to use word-processing, spreadsheet and database management applications to design and implement computer-based solutions to ensure that the public is well informed of the government's new housing initiative; that an efficient record of applicants' information is maintained; applicants' financial income and expense commitments are accurately calculated and assessed and that suitably qualified and approved applicants are duly informed.

Spreadsheet

The Authority of Housing Ministry is a state agency mandated to provide affordable housing for middle and low income earners. The Authority of Housing Ministry invites applications from citizens who are employed and who can demonstrate that they would be able to meet their mortgage payments each month.

Each application must be supported by documents giving details of their financial status. These include the applicant's salary and monthly salary deductions. The amount of money they owed to financial institutions (including: bank payment, credit union payment, and credit card payments) must also be disclosed with supporting verifying documents.

Homes are being built to accommodate **approved applicants** in three communities: **Fitzhope Housing Development (FHD)**, **Hauraruni Housing Development (HHD)** and **La Parfaite Harmonie (LPH)**. Applicants, who are middle and low income earners, must meet a net monthly income (Gross salary – monthly deductions) in order to qualify for consideration in each of the communities. The net salary must be at least: $250,000, $300,000 and $350,000 respectively.

You are required to:

1. Design a spreadsheet that accepts income data from applicants for each housing community. You may accept ten applications for each community. Name the workbook **Application_Data**:
    a. The first worksheet should be named: "**Qualifying Salary**" and should include the housing community's acronym and the qualifying salary as shown below:

Qualifying Salary	
Area	**Cost**
FHD	**$ 350,000.00**
HHD	**$ 300,000.00**
LPH	**$ 250,000.00**

The second worksheet should be named "**Applicant Details**" and should include the applicant's name, housing community acronym, basic salary, bonuses & allowances, gross salary (basic salary + bonuses and allowances), net income (gross salary – salary deductions), qualifying salary and qualifying status.

Bonus and allowances is equated to 2.5% of the basic salary. The salary deduction includes 8% of the basic salary after the first $80,000, plus $10, 000 for insurance. Use functions to determine the **qualifying salary** and **qualifying status**. The **qualifying salary** is found by searching for the housing community in "**worksheet 1**" and returning "**cost"** column. The **qualifying status** will insert the words "Qualified" or "Not Qualified". To be qualified the applicants net salary must be more than or equal to the "**qualifying income**" for the specified housing community.

a) You must enter income data and use appropriate formulae and function to calculate values for each column of the "Applicants Income" worksheet.

	A	B	C	D	E	F	G	H	I	J
2	**First Name**	**Last Name**	**Housing Community**	**Basic Salary**	**Bonuses & allowances**	**Gross Salary**	**Salary Deduction**	**Net income**	**Qualifying Salary**	**Qualifying Status**
3	John	Jones	FHD	XXXXXXX	XXXX	XXXXXXX	XXXX	XXXXXXX	XXXXX	Qualified
4	May	Ally	LPH	XXXXXXX	XXXX	XXXXXXX	XXXX	XXXXXXX	XXXXX	Not Qualified
5	Paul	Browne	HHD	XXXXXXX	XXXX	XXXXXXX	XXXX	XXXXXXX	XXXXX	Not Qualified
6	Geeta	singh	FHD	XXXXXXX	XXXX	XXXXXXX	XXXX	XXXXXXX	XXXXX	Qualified
7	Shivani	Naraine	FHD	XXXXXXX	XXXX	XXXXXXX	XXXX	XXXXXXX	XXXXX	Qualified

b) After the **qualifying status** column, insert information on applicants' loan payments (Bank, credit card and credit union loans), and Monthly payments for the loans (bank payment, credit card payment and credit union payment).

c) Also insert a column for the 'Total Loan Payments'.

d) All candidates took the maximum credit union loan of $30,000.
Most candidates have all three of the loans, while some candidates do not have either a bank loan or a credit card loan.

Loans

i. Repayment on bank loans is calculated as equal monthly payments for 60 months at "**0.006**" interest rate. Bank loan amount must be between 400,000 and 1,000,000).

ii. Repayment on credit card loans is calculated as equal monthly payments for 18 months at "0.004"% interest rate. Credit card loan amount must be between 20,000 and 50,000.

iii. All candidates are given a period of two and a half years to repay credit union loans at an interest rate of 4% per annum.

e) You must enter loan information for each applicant and use appropriate formulae and functions to calculate monthly repayments. Insert a column to calculate each applicant's "**total loan payments**"

After the "**total loan payment**" add a column named "**Balance**", which will show the balance remaining after the total loan repayment is removed from the net income. The values should be centre aligned and formatted to **one decimal place**.

Insert a final column will state whether an applicant has been "approved" or "Not Approved" for housing allocation. An application is approved if the "**Balance**" is equal to or exceeds one half of Net Income.

Total Loan Amounts			Monthly Payments				Approval Status	
Bank	Credit Card	Credit Union	Bank Payment	Credit card Payment	Credit Union Payment	Total Loan Payments	Balance	Approval Status
xxxxxxxx	xxxxxxxx	$30,000.00	xxxxxxxx	xxxxxxxx	xxxxxxxx	xxxxxxxx	xxxxxxxx	Approved
xxxxxxxx	xxxxxxxx	$30,000.00	xxxxxxxx	xxxxxxxx	xxxxxxxx	xxxxxxxx	xxxxxxxx	Not approved
xxxxxxxx	xxxxxxxx	$30,000.00	xxxxxxxx	xxxxxxxx	xxxxxxxx	xxxxxxxx	xxxxxxxx	Approved
xxxxxxxx	xxxxxxxx	$30,000.00	xxxxxxxx	xxxxxxxx	xxxxxxxx	xxxxxxxx	xxxxxxxx	Not approved

Create a suitable chart that compares each Gross Income, and Net income of applicants from the housing communities. Give the chart an appropriate chart title, along with horizontal and vertical axis titles. Place the chart on a new worksheet named **chart 1**.

2. a) For each housing community, sort the applicants' last name in ascending order and their "Net Income" by descending order (provide evidence of the sort).

b) In a new worksheet, create a table that list a summary of the housing communities by basic salary, Gross salary, Net Income, salary deductions and total loan payments.

c) Filter the applicant's income workbook to show all applicants with an income of over $700,000, **provide evidence of the criteria used to filter the data**.

Evidence should be provided for the following (a screen shot snipping should be taken and placed on a separated worksheet named "Evidence" and labelled correctly):

1. Sorting of the data
2. Filtering of the data

Database Management

1. Information must be maintained on all applicants for housing units. Using information from your spreadsheet, you are required to design and populate a database with applicant data. You may accept at least thirty but no more than forty-five applicants.

2. Required personal information on applicants should include Applicant ID, title name, gender, address line 1, address line 2, date of birth, marital status, number of dependants, and phone number and housing community (use acronyms).

3. Applicant ID is a five-digit number in the format XXX026 where XXX is any number between 100 and 999. Applicants must be at least twenty years old but not older than forty-five.

4. Financial data on each applicant is also required. These include net income, total monthly expenses, total monthly repayment, and whether application is approved or not.

You are required to:

i. Design and create database tables (files) to meet the required criteria.
ii. Appropriate data types are to be used when designing the structure of each table.

Your database should respond to the following queries and report.

a) List the last name, address, phone number, net income and number of dependants of all applicants who listed La Parfaite Harmonie as their choice for housing allocation. Name this query LPH_Applicants_Query.

b) List the name, age and total monthly expenses of all applicants between the ages of twenty and thirty whose total monthly loan repayment exceed one thousand five hundred dollars. Name this query LoanPayment_0f_20to30_Query.

c) The AHM wants to know the disposable income of La Parfaite Harmonie's applicants. Perform a query which subtracts the sum of their total loan payments and total expenses from their net income. Display the names loan payment, expenses and disposable income of each La Parfaite Harmonie applicant. Name this query Balance.

d) Perform a query to add a further ten percent to the salary of all approved applicants. The query should display applicant's name, salary and Increased Salary. Name this query Updated_Salary_Query.

e) Prepare a report which lists the names of all applicants whose application have been approved for each housing unit allocation. The report must show each applicant's title, name, applicantID, address, and housing community (acronyms). The report should be sorted by community then by last name. For each grouping level the report must indicate the applicants' average net income. A grand total of income and loan payments and expenses should be indicated at the end of the report. The first line of the report title should be Approved Applicants.

Wordprocessing

1. You are to design a fillable application form on which applicants may apply to the Authority of Housing Ministry for housing allocation. The application form must be designed on letter size paper (8 ½" x 11") using suitable features of a Wordprocessing application. Applicants, name, age, address, DOB, gender, picture, housing community applied for, as well as the applicant's financial information. There should be a space for the applicant's signature. The left and right margins of the form must be 0.8" The top and bottom margins must be set to 1.0"

2. Individuals whose applications have been approved must be duly informed by the Authority of Housing Ministry. Write a letter which congratulates successful applicants and informs that their applications have been approved. The paragraph should contain the statement "Please find a listing of all applicants whose applications have been approved. Please check to ensure that your name is listed:"

   Insert the report named 'Approved Applicants' from the database.

   A final paragraph should give information about the proposed dates for the formal opening of the housing communities.

   Using the mail merge feature of your Wordprocessing program, and data from your database, this letter must be sent from the office of the Chief Executive Officer of the AHM and addressed to the first four and to the last mentioned approved applicant.

Problem-Solving

1. Develop a pseudocode to determine whether an applicant is approved for one of the following housing community: La Parfaite Harmonie (LPH – cost: $150,500), Hauraruni Housing Development (HHD – cost: $440,000) and Fitzhope Housing Development (FHD – cost: $700,500).

    a) The pseudocode should accept the housing community acronym, name of the applicant and their gross salary and total salary deductions. The net salary should be calculated (gross income-salary deductions). An applicant qualifies for the selected housing community if their net salary is more than or equal to 75% of the cost for the housing community.
    b) The algorithm should display "Applicant Qualify" or "Applicant Is Not Qualified), if an applicant is not qualified, the algorithm should then accept information for another applicant.
    c) The algorithm should then ask the applicant for their total utility expense and total living expense. It should then determine whether the applicant is approved. Approval is granted if the applicant's savings is more than or half of their utility and living expenses combined (savings = net salary – (total utility expense + living expense).
    d) The algorithm should display the applicant's name, the cost and full name of the housing community and the applicant's Approval Status ("Approved" or "Not Approved").
    e) The algorithm should then accept information for another applicant if there are other applicants or it should end the program.

2. Design and execute a trace table that accepts data for applicants of one housing community (use acronyms). The table should accept the net salary, total expense, and total repayment amount for each applicant. The balance should be determined. The approval status should be determined (applicant is approved if balance >= half of net income). The number of applicants as well as the number of approved applicants should be counted. In addition, the average net income and average balance of approved applicants should be determined.

Program Implementation

3. Using the programming language Pascal, write program code to implement the algorithm in above.